Carb Burn

Estimating Carbohydrate Oxidation from Cycling Power, LT1 and FTP

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Developed with assistance from Claude (Anthropic, Opus 4.8)

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Abstract

This paper documents the calculations used by the **Carb Burn** data field to estimate whole-body carbohydrate (CHO) oxidation during cycling in real time, using only the head unit's power signal together with two athlete-specific thresholds: the first lactate / aerobic threshold (LT1) and Functional Threshold Power (FTP). The method proceeds in three steps: (i) mechanical power is converted to metabolic energy expenditure through a gross-efficiency factor; (ii) the instantaneous fraction of energy supplied by carbohydrate is estimated with a logistic "crossover" function anchored to LT1 and FTP; and (iii) carbohydrate energy is integrated over time and converted to grams. We also describe the FTP-only fallback, the smoothed grams-per-hour rate, the per-substrate gram totals and the fat-max power, and two uses of body weight: reconciliation against the head unit's own calorie total and an estimate of glycogen-store depletion. All modelling choices are grounded in the exercise-physiology literature, and the known limitations — principally the large inter-individual variability in substrate use — are stated explicitly.

1. Purpose and scope

Direct measurement of substrate oxidation requires indirect calorimetry (respiratory gas exchange), which is impractical on the bike. The gold-standard stoichiometric equations relate the respiratory exchange ratio (RER) to carbohydrate and fat oxidation rates [3]. The Carb Burn field instead produces a *field estimate* from data that a cycling head unit already records — power — combined with thresholds the athlete can supply from testing. It is intended for pacing, fuelling guidance, and post-ride review, not as a laboratory substitute.

2. Notation

Symbol	Meaning	Units
P	Instantaneous mechanical power (from power meter)	W
GE	Gross efficiency (user setting, default 0.21)	–
LT1	First lactate / aerobic threshold power	W
FTP	Functional Threshold Power	W
f(P)	Fraction of energy from carbohydrate at power P	0–1
E	Cumulative metabolic energy expenditure	kcal

m	Cumulative carbohydrate mass oxidised	g
BW	Body weight (user setting)	kg

3. Step 1 — Mechanical power to metabolic energy

A calibrated power meter measures external mechanical work directly. Metabolic energy expenditure is recovered by dividing mechanical power by gross efficiency, the ratio of mechanical work performed to metabolic energy consumed:

$$P_{met} = P / GE$$

Multiplying accumulated mechanical work by a gross-efficiency factor is a validated way to estimate energy expenditure from a power meter: group-level estimates agree with indirect calorimetry to within ~1%, although individual error can reach ~11% when a population-mean efficiency is used instead of the rider's own value [1]. Using an individualised, power-specific gross efficiency significantly improves accuracy [2]. Gross efficiency in trained cyclists typically falls between 19% and 24%; the field exposes GE as a user setting (default 21%) so it can be personalised. The metabolic energy accumulated over a sampling interval Δt is

$$\Delta E = (P / GE) \cdot \Delta t / 4184 \quad [kcal]$$

where 4184 J = 1 kcal and Δt is derived from the activity timer so that paused time is never accumulated.

4. Step 2 — Carbohydrate fraction from power

Fuel selection shifts predictably with exercise intensity. At low intensity fat predominates; as intensity rises the carbohydrate share increases through a "crossover point", beyond which carbohydrate is the dominant fuel [6, 7]. The fraction of muscle fuel from carbohydrate is directly predicted by power output [5], and the carbohydrate–power relationship has been formalised as the "carbohydrate cost of the watt" in a cohort of 6465 subjects [4]. Because the transition is smooth and saturating, we model the carbohydrate energy fraction as a logistic function of power:

$$f(P) = 1 / (1 + e^{-k(P - P_{50})})$$

where P_{50} is the power at which carbohydrate supplies 50% of energy (the crossover point) and k sets the steepness. The two free constants are fixed by anchoring the curve to the athlete's own thresholds. LT1 marks the aerobic threshold, where fat oxidation is still substantial but carbohydrate use is climbing; FTP sits near the upper boundary of the heavy-intensity domain, where carbohydrate dominates [9, 10]. We therefore impose:

$$f(LT1) = 0.35 \qquad f(FTP) = 0.85$$

Taking the logit (the inverse of the logistic), $\ln[f / (1 - f)] = k(P - P_{50})$, and substituting the two anchor points:

$$\ln(0.35 / 0.65) = -0.6190 \qquad \ln(0.85 / 0.15) = +1.7346$$

Subtracting the first relation from the second eliminates P_{50} and yields the steepness; back-substitution gives the crossover power:

$$k = 2.3536 / (FTP - LT1)$$

$$P_{50} = LT1 + 0.2631 \cdot (FTP - LT1)$$

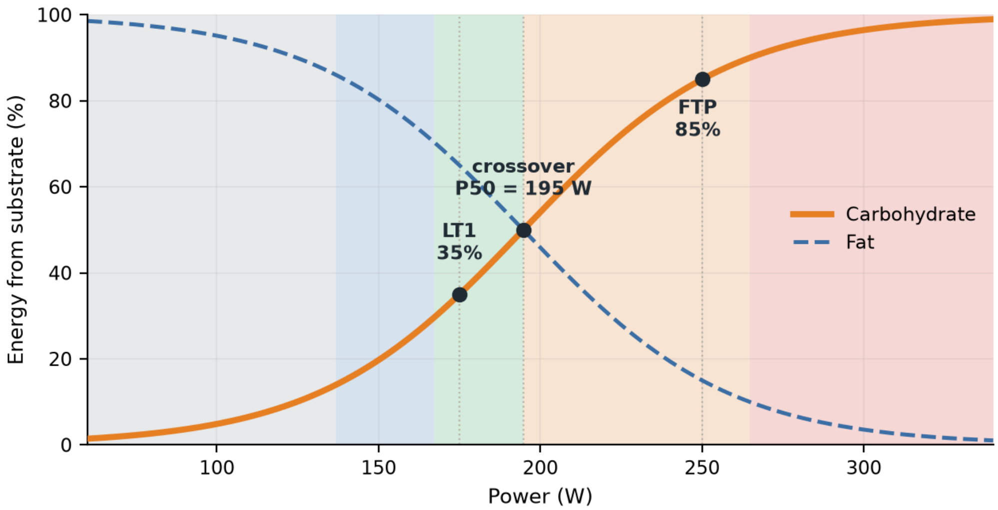

Figure 1. *Modelled substrate split for a generic sample rider (FTP = 250 W, LT1 estimated at 175 W, GE = 0.21). The carbohydrate fraction f(P) follows the logistic curve anchored at 35% at LT1 and 85% at FTP; fat supplies the remainder. The crossover point (50% carbohydrate) falls at* $P_{50} \approx 195$ *W. Shaded bands mark the app's power-based pacing colour zones (grey → blue → green → orange → red).*

Thus the crossover point sits roughly one-quarter of the way from LT1 to FTP, consistent with reported crossover intensities lying above the aerobic threshold but below maximal effort [6, 7]. This threshold-anchoring is conceptually aligned with lactate-based substrate models, in which the relative carbohydrate rate is a sigmoid function of blood lactate concentration [8] — LT1 and FTP being, in effect, lactate landmarks. The exponent is clamped to ±30 to prevent numerical overflow at extreme power.

5. Step 3 — Carbohydrate energy to grams

The instantaneous carbohydrate energy increment is the metabolic energy increment scaled by the carbohydrate fraction. Summing across the ride and dividing by the energy density of carbohydrate (4.0 kcal per gram [3]) gives the cumulative carbohydrate mass:

$$m = (1 / 4.0) \cdot \Sigma f(P) \cdot (P / GE) \cdot \Delta t / 4184 \quad [g]$$

The session-average carbohydrate share reported as "% CARB" is the ratio of cumulative carbohydrate energy to cumulative total energy, $E_{CHO} / E_{total} \times 100$.

6. FTP-only fallback (no LT1 test)

Many riders know their FTP but have never had a formal LT1 / aerobic-threshold test. When LT1 is not supplied, it is estimated as a fixed fraction of FTP:

$$LT1 \approx 0.70 \cdot FTP$$

The aerobic threshold in trained cyclists commonly falls near two-thirds to three-quarters of FTP, placing it within the moderate-intensity domain below the maximal lactate steady state [9, 10]. The same logistic machinery of Section 4 is then applied unchanged, so the tool degrades gracefully to an FTP-only model. A guard enforces FTP > LT1 to keep the steepness finite.

7. Oxidation rates, cumulative grams, and fat max

The instantaneous carbohydrate burn rate is the per-second carbohydrate energy expressed per hour and converted to grams:

$$r_{inst} = f(P) \cdot (P / GE) / 4184 \cdot 3600 / 4.0 \quad [g/h]$$

Because power is intermittent, the displayed rate is passed through a first-order exponential moving average with smoothing factor α = 0.10 (time constant ≈ 10 s), and decays toward zero while coasting:

$$r_n = r_{n-1} + \alpha \cdot (r_{inst} - r_{n-1})$$

The identical construction gives the fat oxidation rate, using the higher energy density of fat (9 kcal/g [3]) and the complementary fraction 1 − f(P):

$$r_{fat} = (1 - f(P)) \cdot (P / GE) / 4184 \cdot 3600 / 9.0 \quad [g/h]$$

Figure 2 plots both substrate rates against power for the sample rider. Two features are worth noting. First, carbohydrate use climbs steeply and without bound as power rises, whereas fat oxidation rises to a peak ("fat max") in the moderate domain and then falls away — the physiological basis of low-intensity "fat-burning" training. Second, because fat carries more than twice the energy per gram, its contribution in *grams* stays modest even where its energy *share* (Figure 1) is large; carbohydrate dominates the gram tally at any appreciable intensity. By mass the two fuels are equal at a lower power (~170 W here) than the 50% energy crossover of Figure 1 (195 W), precisely because of this density difference.

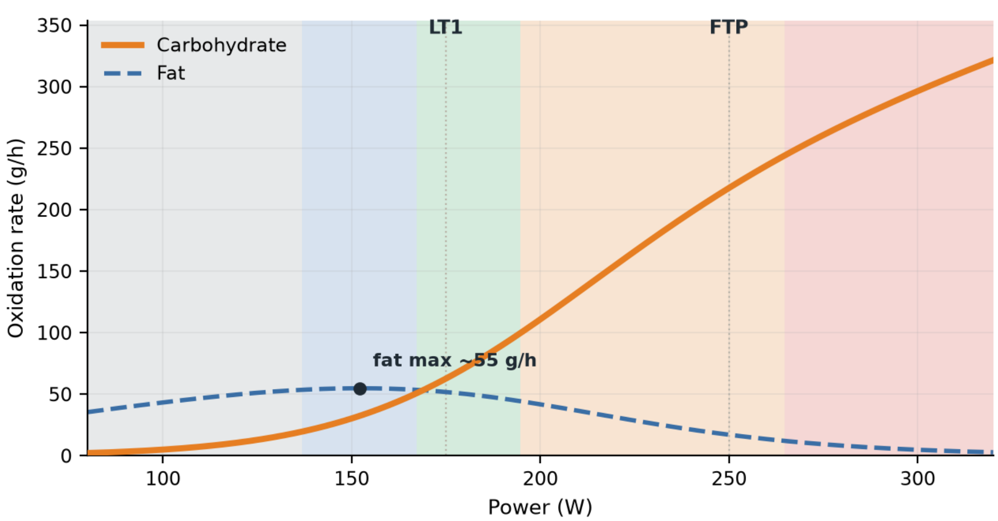

Figure 2. *Modelled carbohydrate and fat oxidation rates (grams per hour) versus power for a generic sample rider (FTP = 250 W, LT1 = 175 W, GE = 0.21). Carbohydrate grams rise steeply with intensity; fat grams peak near the fat-max power and decline. Shaded bands mark the app's pacing colour zones. Values are modelled estimates, not measurements.*

7.1 Cumulative fat mass (FAT)

Fat mass is accumulated exactly as carbohydrate mass (Section 5) but with the complementary fraction 1 − f(P) and the energy density of fat (9 kcal/g). This drives the FAT readout shown on large layouts:

$$m_{fat} = (1 / 9.0) \cdot \Sigma (1 - f(P)) \cdot (P / GE) \cdot \Delta t / 4184 \quad [g]$$

As with the carbohydrate totals, it is scaled by the Garmin reconciliation factor R (Section 8.1) when a device calorie total is available, so carbohydrate grams, fat grams and total energy stay mutually consistent.

7.2 Fat-max power (FATMAX)

The fat-max power is the intensity at which the fat oxidation rate $r_{fat}(P)$ is greatest — the wattage a rider targets to maximise fat use. The constant factors in r_{fat} do not move the location of the peak, so it is enough to maximise g(P) = P · (1 − f(P)). Differentiating the logistic (f' = k f (1 − f)) and setting g'(P) = 0 gives a compact implicit condition:

$$P_{fatmax} \cdot k \cdot f(P_{fatmax}) = 1$$

Because f(P) itself depends on power, there is no closed-form solution; the field solves it once per settings change by scanning power in small (2 W) steps for the peak — cheap, since it runs only when FTP, LT1 or GE change. Under the anchoring of Section 4 the solution lands near 60–64% of FTP across a wide range of thresholds (for example, 152 W for a 250 W FTP with LT1 estimated at 175 W), consistent with reported fat-max intensities sitting in the moderate domain below LT1.

8. Uses of body weight

The power-based model is deliberately weight-independent: metabolic energy is recovered from power and efficiency without reference to body mass. Body weight is used only for two secondary functions.

8.1 Garmin calorie cross-check

The head unit maintains its own cumulative calorie total (C_{Garmin}), computed using the rider's profile weight. When available, the field rescales only the *magnitude* of its carbohydrate output to agree with that total, using a reconciliation factor:

$$R = C_{Garmin} / E_{model} \quad (R = 1 \text{ if unavailable})$$

The reconciled grams and rate are m · R and r · R. Crucially, the carbohydrate *fraction* ("% CARB") is invariant under this rescaling, since R cancels in the ratio; only total energy is reconciled, preserving the physiologically meaningful split derived from power and thresholds. Applying an individualised gross efficiency to power is itself the most accurate energy-estimation route [1, 2], so R principally corrects for differences between the user's GE setting and the head unit's internal assumptions.

8.2 Glycogen-store depletion

Total body glycogen scales with body mass. Taking a representative store of about 8 g per kilogram of body mass, the field expresses cumulative carbohydrate oxidised as a share of estimated stores:

$$D = m / (BW \cdot 8) \times 100 \quad [\%]$$

This is an intentionally simple depletion index: it ignores carbohydrate ingested during the ride and inter-individual variation in storage capacity, and is offered as context rather than a precise balance. Endurance athletes deliberately keep most training below threshold precisely to protect glycogen homeostasis [9], which motivates surfacing this quantity to the rider.

9. Worked example

Consider a rider with FTP = 250 W, no LT1 test, GE = 0.21, holding P = 200 W.

Quantity	Expression	Value
Estimated LT1	*0.70 x 250*	175 W
Steepness k	*2.3536 / (250 - 175)*	0.03138 /W
Crossover P50	*175 + 0.2631 x 75*	194.7 W
CHO fraction f(200)	*1 / (1 + e^-0.03138(200-194.7))*	0.541
Metabolic power	*200 / 0.21*	952 W
Energy rate	*952 / 4184 x 3600*	819 kcal/h
Carb burn rate	*0.541 x 819 / 4.0*	**~111 g/h**

Sustained for one hour this rider would oxidise roughly 111 g of carbohydrate with about 54% of energy drawn from carbohydrate — a plausible tempo-intensity figure.

10. Assumptions and limitations

- **Population-calibrated, not measured.** The carbohydrate/fat split is modelled, not sampled from gas exchange. True substrate use shows large inter-individual variability that intensity alone does not explain [3, 11].
- **Diet and training status shift the crossover.** Low-carbohydrate/ketogenic adaptation can move the crossover to a much higher intensity and raise peak fat oxidation substantially, which the fixed anchors do not capture [6].
- **Sex and body composition matter.** Women tend to oxidise relatively more fat and men more carbohydrate at matched intensities [11]; the model uses a single generic curve.
- **Gross efficiency is assumed constant.** In reality GE varies mildly with intensity, cadence and fatigue [1, 2]; a single user value is a simplification.
- **Stoichiometric assumptions.** The 4.0 kcal/g carbohydrate density and the neglect of the bicarbonate pool at very high intensity follow the standard calorimetric equations and their known caveats [3].
- **Glycogen index is indicative only.** The 8 g/kg store and the omission of on-bike carbohydrate intake make the depletion figure a rough guide.

For individual accuracy, the gross-efficiency value and the two anchor fractions (35% / 85%) can be calibrated against a laboratory metabolic (RER) test; both are exposed in the source for tuning.

References

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Prepared as engineering documentation for the Carb Burn Connect IQ data field by Stephen Cieply, PhD, with development assistance from Claude (Anthropic). The model provides field estimates for training guidance and is not a medical or laboratory instrument. Literature located via peer-reviewed database search; readers should consult the primary sources for full methods and context.